Roll No. | | | | | | | | | | |

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharma (Sem.-6)

PHARMACEUTICAL BIOTECHNOLOGY-THEORY

Subject Code : BP-605T

M.Code : 77990

Date of Examination : 27-05-2023

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. **Explain the following terms in brief :**

   a. Biotechnology

   b. Genetic Engineering

   c. Enzymes

   d. Lipase

   e. Cloning

   f. ELISA

   g. Plasma

   Distinguish between :

   h. Vaccine and antibiotics

   i. Active and Passive immunity

   j. Toxin and Toxoid.

SECTION B

2. Enumerate general methods for the production of vaccines.

3. Explain production of

   a. Toxoid

   b. Serum immune blood derivatives

4. Write short note on

   a. Hepatitis B vaccine

   b. Vitamin B 12

SECTION C

5. Explain the methods of enzyme immobilization.

6. What are biosensors. Explain its working.

7. Brief the application of recombinant DNA technology for the production of interferon.

8. Enumerate designing of thermodenaturation processes.

9. Give construction and working including controlling of parameters of a single stage fermenter.

10. Enumerate the method of collection, processing and storage of whole human blood.

11. Name various immune testing techniques. Explain any one with example.

12. What is PCR? Give its applications.

13. Write short note on

   a. Hybridoma technology

   b. Types of mutation.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

July - 2021